MIL-S-2412D
1 September 1978
SUPERSEDING
MIL-S-2412C
8 June 1970

MILITARY SPECIFICATION

SCALE, PLOTTING, COORDINATE:

PLASTIC, GRADUATED IN YARDS AND METERS IN

1:25000 AND 1:50000 SCALES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a flat, plastic, L-shaped, coordinate scale.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-535	- Plastic Sheet (Sheeting); Plastic Strips; Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The scale shall be flat, L-shaped, laminated plastic, graduated in meters on one side and yards on the other side in 1:25000 and 1:50000 scales as shown on Figures 1 and 2, and as specified herein.

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3.2 First article (first produced scale). The contractor shall furnish one or more scales as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce scales that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval of the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Surface and edges. The surface shall be smooth and the edges shall be square and straight to within 0.003 inch (0.08 mm) of the precision flat (see 4.5.2.1). The outside edges shall be perpendicular to within 0.014 inch (0.36 mm) and the inside edges shall be perpendicular to within 0.010 inch (0.25 mm). Sharp edges shall be rounded to a radius not to exceed 0.003 inch (0.08 mm) and all corners rounded to the radii shown on Figure 2.

3.5 Graduations, numerals, and lettering. The scale shall be graduated, numbered, and lettered as specified herein and as shown on Figures 1 and 2. The graduations, numerals, and lettering shall be marked on rigid unsupported poly vinyl-chloride sheet conforming to L-P-535, Composition A, Type II, Grade NT, Color White and Opaque. The graduations shall be 0.010 inch, plus or minus 0.003 inch (0.25 mm plus or minus 0.08 mm) in width and shall be spaced equidistant with the accumulative tolerance over the entire scale not to exceed 0.010 inch (0.25 mm). Unless otherwise shown on Figures 1 and 2, the numerals and letters shall be Gothic, 0.063-inch (1.58 mm) high. The vertical and horizontal scale zero shall coincide within plus or minus 0.010 inch (plus or minus 0.25 mm).

3.6 Marking color. The meter side of the scale shall be marked in red and the yard side of the scale shall be marked in black. Colors shall conform to FED. STD. No. 595, Red color number 31136 and Black color number 37038.

3.7 Laminations. The graduated and marked scale shall be sealed between two transparent cover sheets conforming to L-P-535, Composition B, Type I, Class 2, Color Clear and Transparent. The finished scale after lamination shall be not less than 0.036-inch (0.91 mm) or more than 0.040-inch (1.02 mm) thick.

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3.8 Workmanship. The permanently attached plastic components of the scale shall be sealed in such a manner as not to impair the transparency of the cover sheets or distort the graduations and markings on the covered portion of the scale. The finished scale shall show no cracks or separation of the sheets and shall be free of scratches, bubbles, imbedded particles, and sharp edges and corners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced scale inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced scale inspection.

4.3.1 Examination. The first produced scale shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced scale shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be conducted in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Test.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The scale shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Edges not square and dimensions not within specified limits.
103. Scales not graduated as specified.
104. Lettering not as specified.
105. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Edge straightness test. Place the edges against a precision straightedge, surface plate, bench block, or other precision flat. Measure the points of noncontact between the edge and the precision flat. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.2 Accuracy test. A beam compass which has been certified as accurate by the National Bureau of Standards shall be used to check the scale graduation marks. An accumulative tolerance over the entire scale exceeding 0.010 inch (0.25 mm) shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

106.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
107.	Gross weight exceeds 200 pounds for Level A.
108.	Strapping not as specified for Level A.
109.	Gross weight or size exceeds the limitations of the box for Level B.
110.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each scale shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.2 Level B. Each scale shall be preserved in accordance with MIL-P-116, Method III, by the use of a wrapping, bag, or carton.

5.1.3 Commercial. Each scale shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

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5.2.1 Level A. The scales, preserved as specified in 5.1, shall be packed in quantities not to exceed 200 pounds in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The scales, preserved as specified in 5.1, shall be packed in close-fitting fiberboard boxes conforming to PPP-B-640, Class 2, style optional or PPP-B-636, V3c, style optional. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The scales, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military packaging shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The scale is intended to be used for reading and plotting coordinates on 1:25000 and 1:50000 scale maps, or map substitutes, bearing either a 1000 meter or 1000 yard grid.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced scale, and number of scales required (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When other than Finish B strapping is required (see 5.2.1).

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6.3 First produced scale. Any changes or deviations of production scales from the approved first produced scale during production will be subject to the approval of the contracting officer. Approval of the first produced scale will not relieve the contractor of his obligation to furnish scales conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

Custodian:
Army - ME

Review activity:
DLA - GS

User activities:
Navy - MC, OS, AS, YD

Preparing activity:
Army - ME

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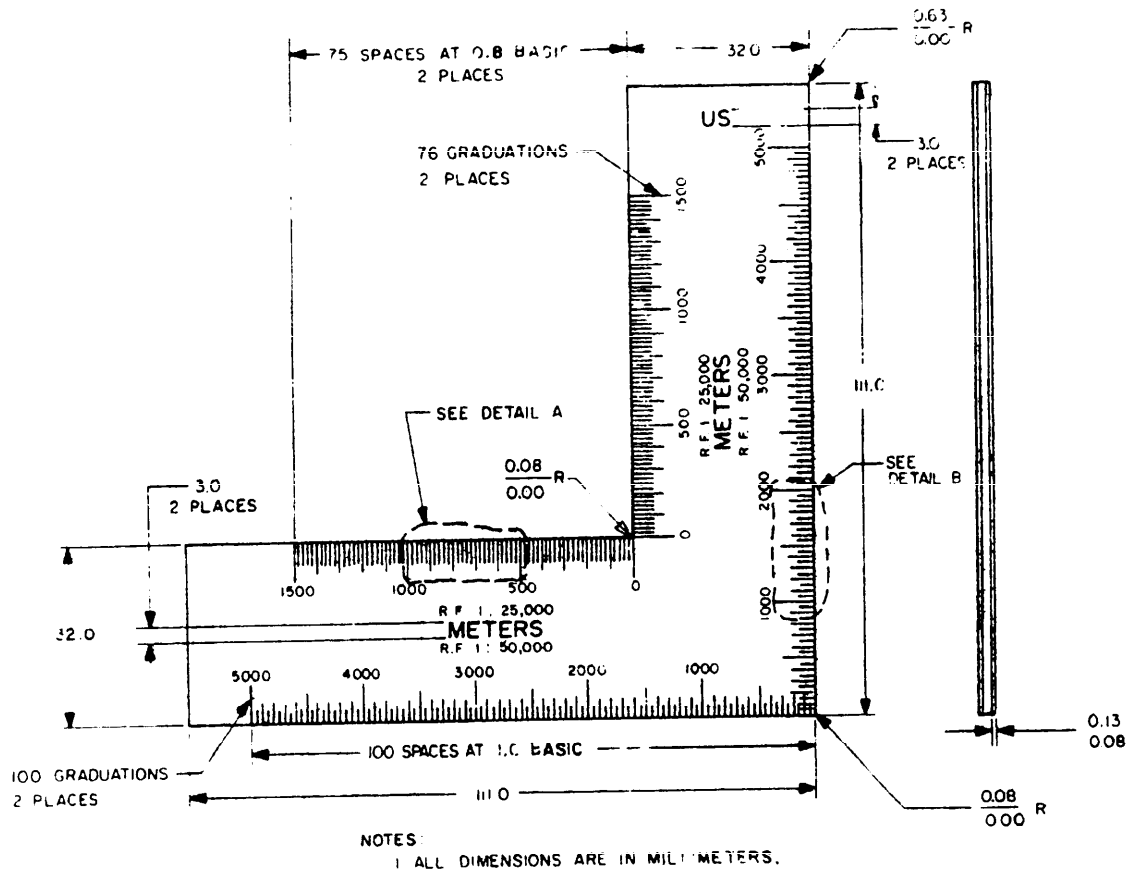
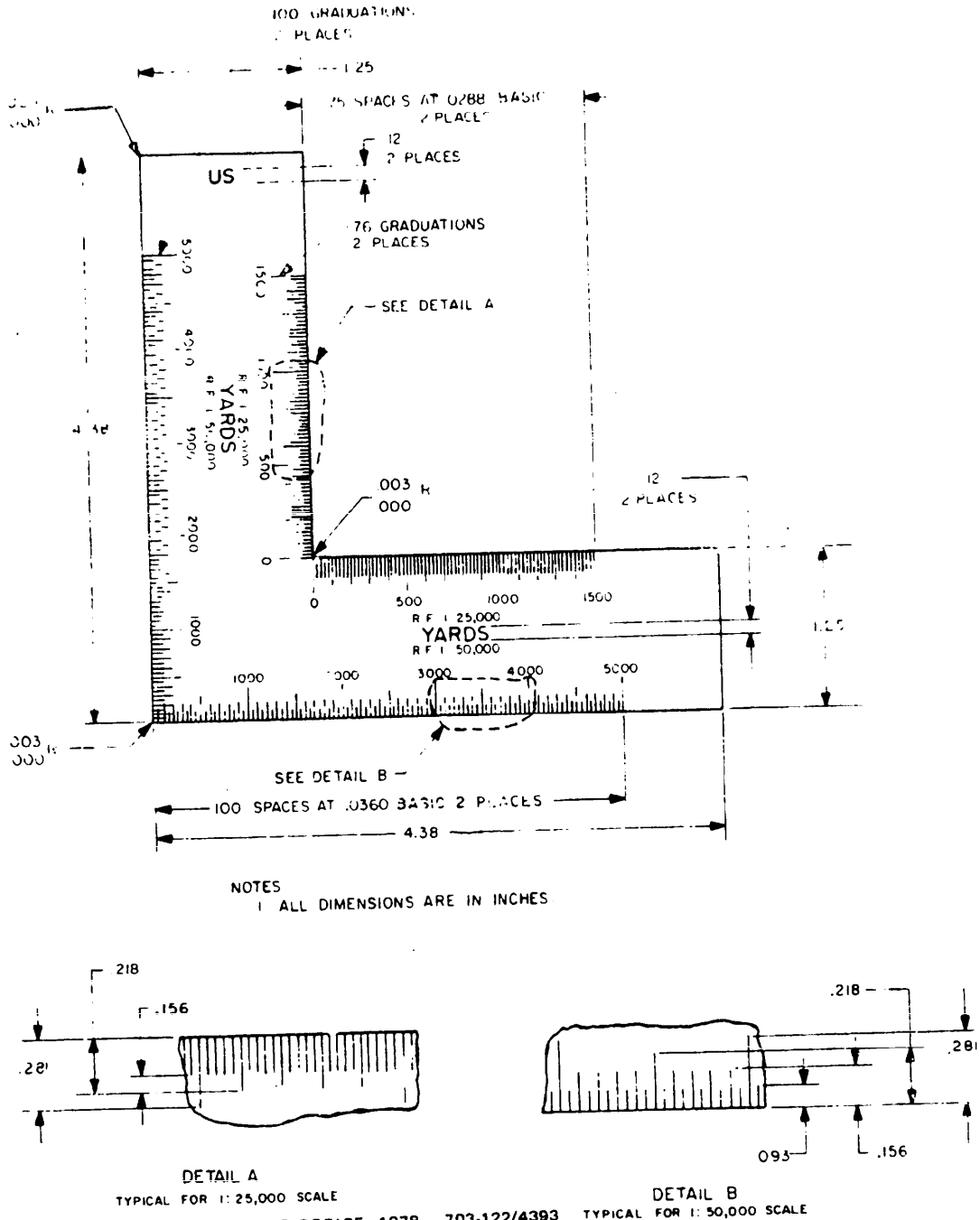


FIGURE 1. Scale plotting graduated in meters.

X-3388

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FIGURE 2. Scale plotting graduated in yards.

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DOCUMENT IDENTIFIER AND TITLE: MIL-S-2412D Scale, Plotting, Coordinate: Plastic, Graduated in Yards & Meters in 1:25000 & 1:50000 Scales

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

I. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

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COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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